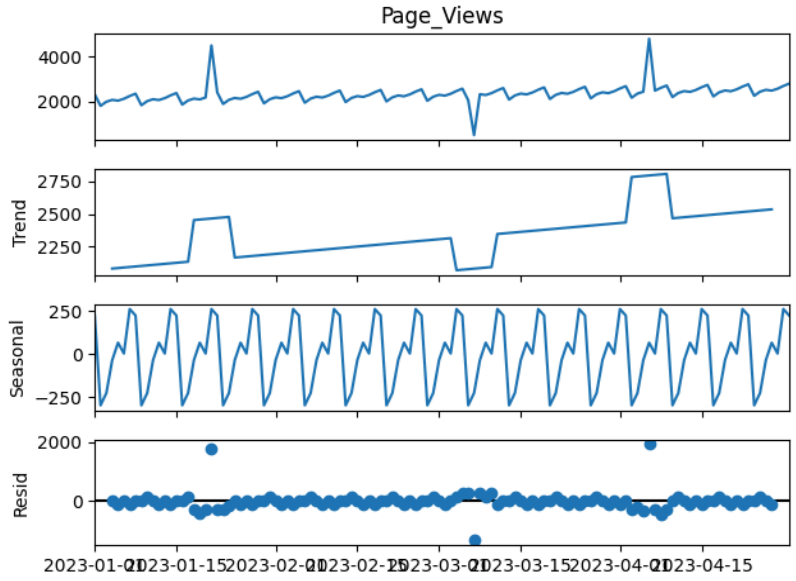
EXPERIMENT NO: 06

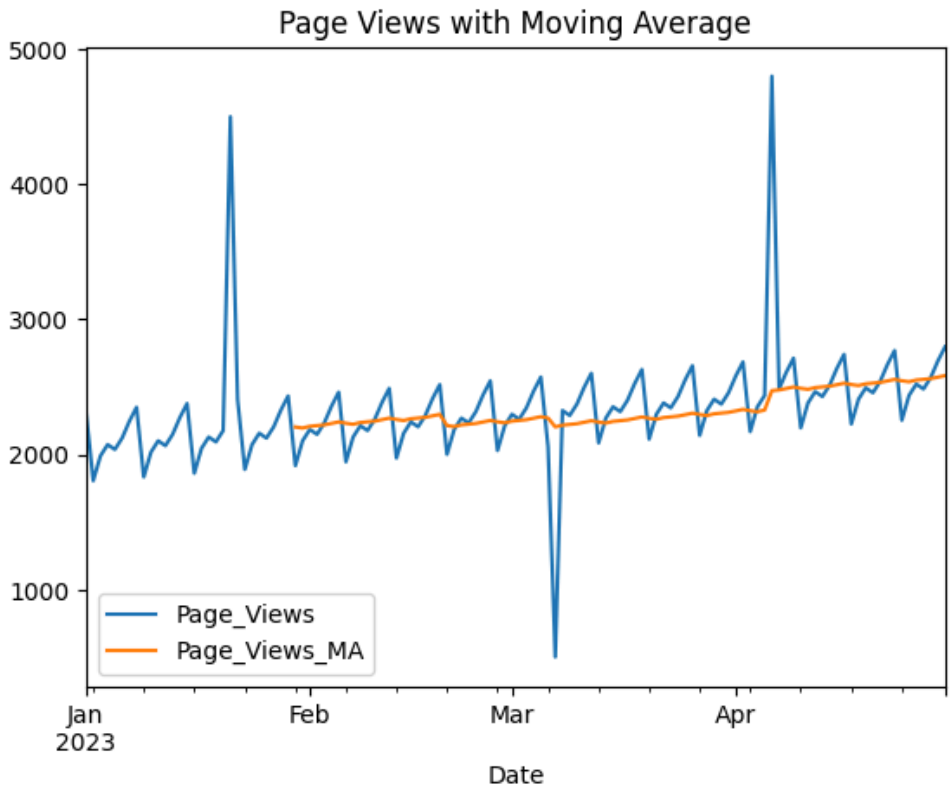
Time Series Analysis of Website Traffic

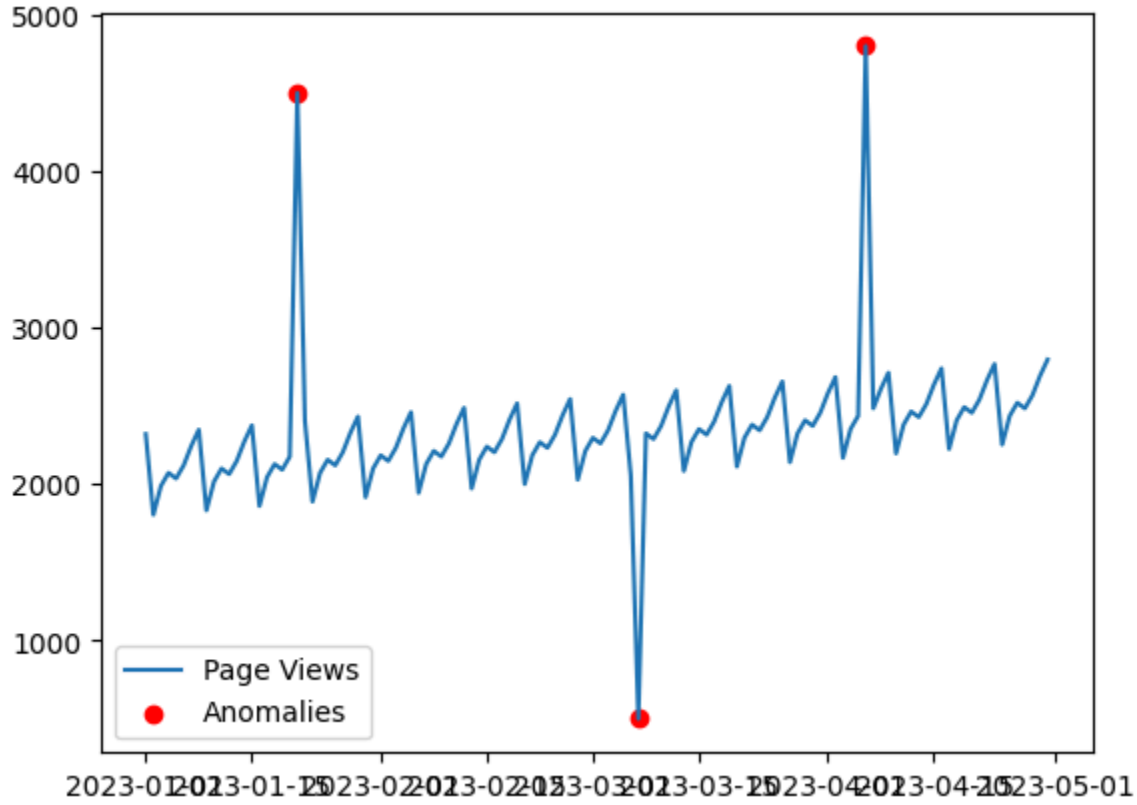
Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
data=pd.read_csv('website_traffic.csv',parse_dates=['Date'],index_col='Dat
e')
decomposition=seasonal_decompose(data['Page_Views'],model='additive')
decomposition.plot()
plt.show()
data['Page_Views_MA']=data['Page_Views'].rolling(window=30).mean()
data[['Page_Views','Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
data['Page_Views_Z']=(data['Page_Views']-data['Page_Views'].mean())/data['
Page_Views'].std()
anomalies=data[data['Page_Views_Z'].abs()>3]
plt.plot(data.index,data['Page_Views'],label='Page Views')
plt.scatter(anomalies.index,anomalies['Page_Views'],color='red',label='Ano
malies')
plt.legend()
plt.show()
```

Output:




Page Views with Moving Average
5000
4000
3000
2000
1000
Page_Views
Page_Views_MA
Jan
2023
Feb
Mar
Apr
Date


5000
4000
3000
2000
1000
Page Views
Anomalies
2023-01-01
2023-01-15
2023-02-01
2023-02-15
2023-03-01
2023-03-15
2023-04-01
2023-04-15
2023-05-01